Brain Tumour Detection using Deep Learning From MRI Scans With Clinical Advice

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ABSTRACT

Brain tumours represent complex and potentially life-threatening neurological conditions that disrupt normal brain function and significantly influence a patient's physical health, cognitive performance, and emotional well-being. Because of their critical nature, early identification, precise localization, and accurate stage determination are essential for improving prognosis and guiding effective therapeutic strategies. This study introduces a transparent and intelligent deep learning framework designed for automatic brain tumour detection and stage classification using Magnetic Resonance Imaging (MRI) data. The proposed model is built upon Convolutional Neural Networks (CNNs), which are specialized architectures capable of extracting intricate spatial features from medical images. By learning meaningful patterns from MRI scans, the system accurately detects abnormal tissue growth and categorizes tumours according to their progression stage with strong predictive performance. To ensure interpretability and foster clinical confidence, the framework integrates Gradient-weighted Class Activation Mapping (Grad-CAM), which visually emphasizes critical regions within the MRI images that contribute most to the model's decisions. This visualization enables healthcare professionals to clearly

observe tumour boundaries, size variations, and affected regions, thereby supporting transparent decision-making. In addition to diagnostic capabilities, the system incorporates a stage-aware clinical recommendation component that suggests potential treatment pathways aligned with tumour severity. To enhance accessibility and usability, a voice-driven interface powered by artificial intelligence is included, allowing users to interact with the system through spoken commands and receive real-time medical guidance. Experimental validation demonstrates high levels of accuracy, stability, and reliability. Overall, this integrated framework offers an interpretable, efficient, and technologically advanced solution for intelligent brain tumour diagnosis and personalized treatment planning in modern healthcare environments.

Keywords: Brain tumour identification, Deep learning techniques, Convolutional Neural Networks (CNN), Grad-CAM visualisation.

1 . Introduction

A sudden cluster of unusual cells shows up inside the skull or nearby nerve areas, forming what doctors call a brain tumour. When regular cell reproduction goes off track, extra tissue builds up - this lump may disrupt how nerves work. Some start right where they're found; others travel from distant cancer sites through the body. Growth type depends on source: one kind begins in the brain itself, while the second arrives after moving from elsewhere.

Some brain growths fall into two groups: those that aren't cancer, called benign, others that are, known as malignant. Even if slow-moving, non-cancer types press on areas inside the head where room is tight, creating big problems. Fast-growing ones tend to push into healthy tissue nearby instead of staying put. Glioblastoma stands out because it moves quickly through brain matter, needing urgent care right away. Treatment must start fast when this type shows up.

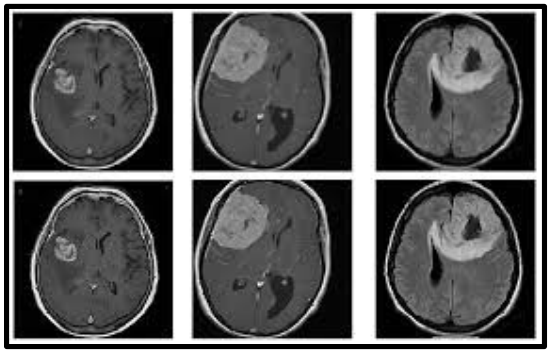


Figure 1.1

Globally, tumours of the brain and spinal cord weigh heavily on public health, says WHO. Depending on where they grow, how big they get, or what kind they are, symptoms shift. Headaches that won't fade often show up first. Seizures might follow without warning. Trouble recalling things pops up in some cases. Eyesight gets blurry for others. Stomach discomfort joins the mix more than expected. Speaking becomes clumsy at times. Balance sometimes falters mid-step.

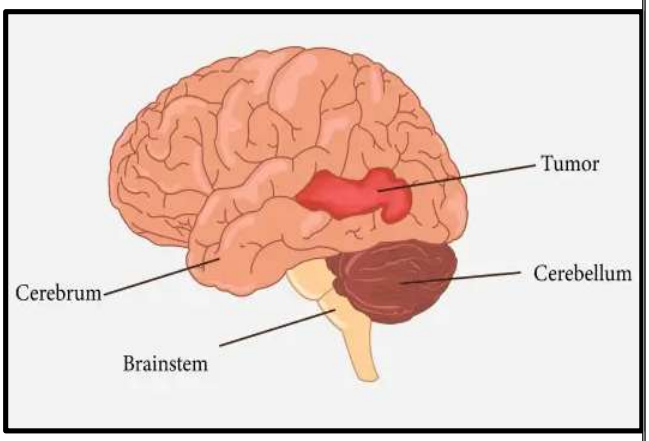


Figure 1.1

Getting a diagnosis usually means using high-tech scans like MRI or CT. What makes MRI stand out is how clearly it shows the brain's softer parts, helping pinpoint where a tumour sits. Spotting things early turns out to matter a lot when it comes to how well someone does during treatment.

Nowadays machines help doctors spot brain growths better. Depending on how far it has spread, what shape it is, and how sick the person feels, different paths open up - cutting it out, zapping it, poisoning fast cells, aiming at weak spots, or mixing methods. As computers learn to see patterns faster, tools built on smart algorithms slowly become part of daily hospital work.

1.1 Importance of Early Detection:

Spotting brain tumours sooner helps people live longer and feel better day to day. Since

the brain runs key tasks like moving, thinking, talking, seeing, and balance, even tiny unusual growths can throw off how nerves work. When found early, treatment can begin quickly - stopping spread and preventing lasting harm to nearby areas of the brain.

Early detection makes treatments work better, while also reducing how much they invade the body. Smaller, contained growths usually allow surgery to succeed more easily than larger ones spread out. Radiation and chemo tend to hit harder against tumours caught sooner, which lifts recovery odds. If diagnosis waits too long, growth spreads might raise brain pressure, bringing serious nerve problems along.

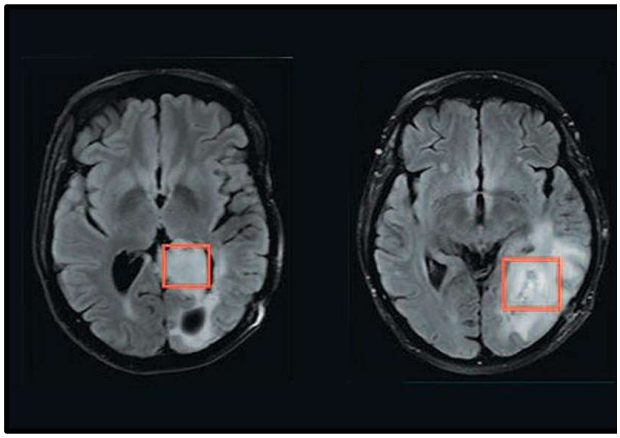


Figure 1.3

One report from the World Health Organization shows spotting symptoms sooner cuts deaths from brain and spinal cord tumours. Because doctors see things earlier, they can better judge how far a growth has spread. That clarity shapes what care works best while giving clearer hints about recovery chances.

Spotting things early often means fewer lasting problems. When a growth gets caught before it reaches vital parts of the brain, thinking skills stay sharper, talking stays easier, movement remains smoother. That shift doesn't just change recovery paths - it eases pressure at home, inside hearts, inside budgets too. Fewer crises pile up when timing leans forward.

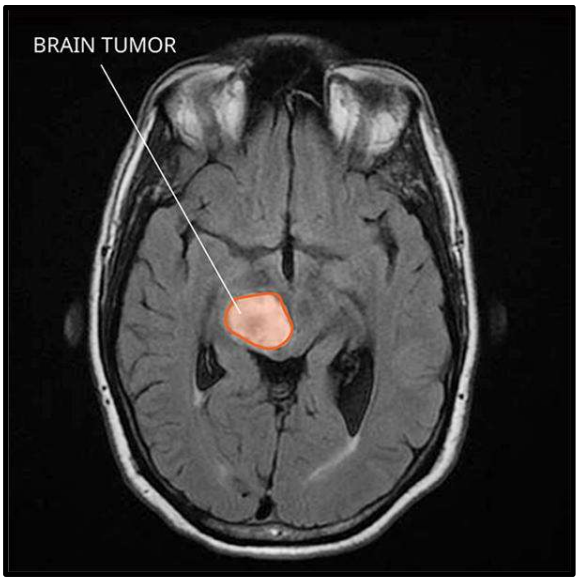


Figure 1.4

Now machines spot tiny changes in scans faster than human eyes. Because AI steps into radiology, small signs hiding in images get noticed earlier. When tech highlights what might slip past a doctor, treatment paths open sooner. So care grows sharper when smart tools join the exam process. Medical choices improve once hidden clues come to light through digital help.

1.2 Problems in Manual Diagnosis:

Looking closely at MRI or CT scans, doctors spot unusual growths to diagnose brain tumours by hand. Despite their skill, experts spend long hours reviewing each image, opening room for personal judgment. Tiny differences - how a mass curves, grows, or feels in the scan - affect

what they see. Because of that, one reader might differ from another without clear agreement.

It often happens that one expert sees a detail another misses when reading scans by hand. Experience shapes how each radiologist reacts to an MRI image, so opinions shift from person to person. Because of this shifting perspective, spotting growths, defining edges, or deciding severity might turn out uneven. Early signs sometimes fade into the background, particularly when changes are faint and barely there.

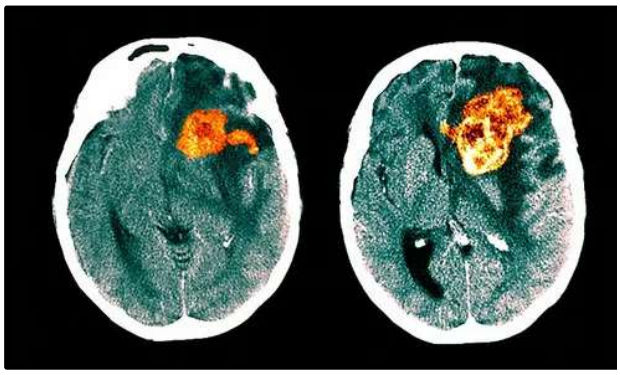


Figure 1.5

Heavy workloads create real challenges. When clinics pile on scans, doctors face long hours staring at images. Tired eyes miss details - timing squeezes only make it worse. Mistakes slip in more easily under that kind of strain. Global health groups like the WHO point out how delays hurt recovery chances. Clearer reads happen when pace eases, they note across research papers and safety rules.

Outlining tumour zones by hand? Tough work. Sharp borders matter yet they often fade into nearby tissue, making edges hard to catch. Tricky shapes make it worse. Deep spots in the brain add another layer, especially where every millimetre counts. Precision slips when anatomy crowds the view.

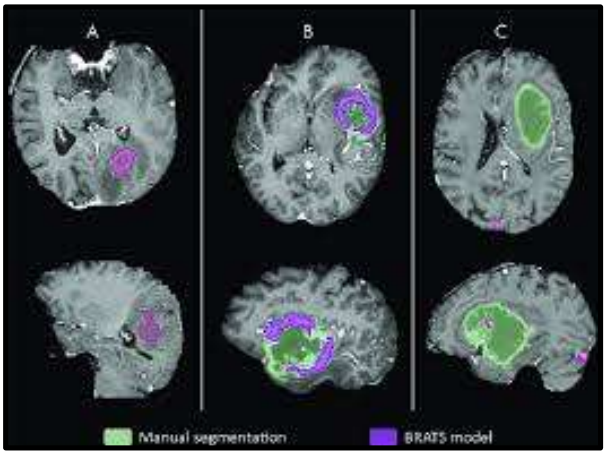


Figure 1.6

Few people notice how tricky it is to spot deep patterns in detailed brain scans just by looking. Hidden signs of cancer growth often slip past even trained eyes. Because of this, depending only on human review can miss subtle warnings at the start. Tough details hide where vision fails.

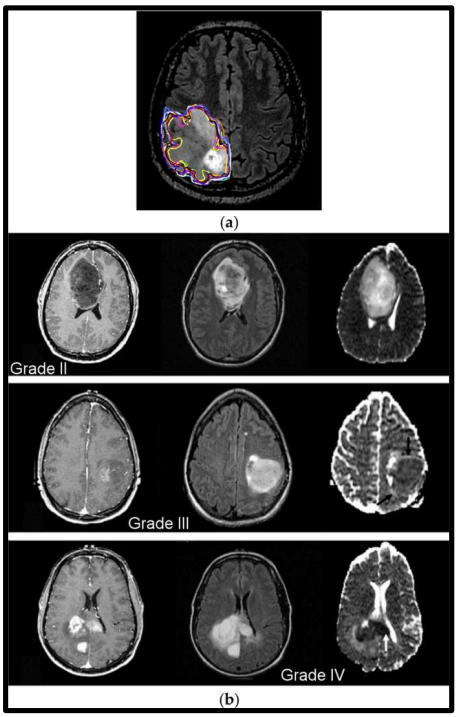
Not every tool fits the moment yet smart automation steps in where speed matters, offering steady readings that match a doctor's judgment without slowing them down.

1.3 My Contribution:

A fresh look at spotting brain tumours comes through a smart setup trained on MRI scans - accuracy stands out, yet clarity matters just as much. Instead of working like a black box, this method shows its reasoning, making results easier to trust. Built for real medical use, it guides decisions without hiding how it arrives at them. What sets it apart? Not just performance, but showing why each result makes sense.

Right off, a setup built on CNNs grabs key details from MRI images without help. Instead of hand-checking each scan, the method spots tricky visual structures and tells healthy tissue apart from flawed ones. Not just spotting issues, it groups tumor stages too - adding depth to how doctors evaluate cases.

Now here comes a shift - instead of hiding how decisions form, this work brings in Grad-CAM to expose them. What does it does? It lights up key spots in brain scans that the system pays attention to. Picture warm glows around tumors, showing exactly where the model looks. That kind of clarity helps doctors see why a result was reached. Trust grows when guesses turn into visible reasoning. Suddenly, the machine isn't just deciding - it's pointing.



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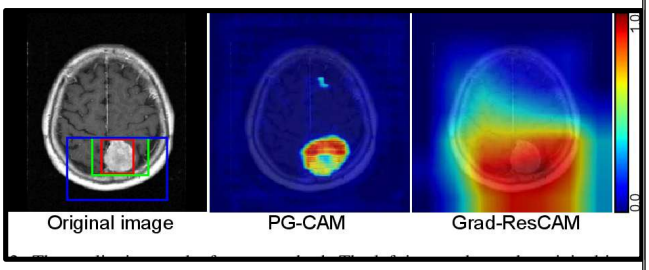

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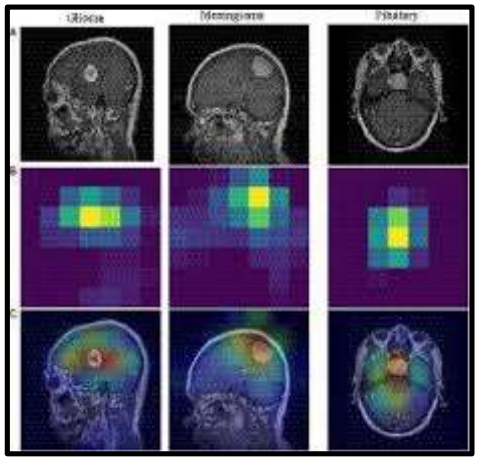


Figure 1.9

What stands out is how the tool adds a smart suggestion layer tuned to cancer stages. When it spots what phase the tumor is in, advice about next steps shows up - matched to how serious things are. Decisions get easier because results link straight into care choices. Jumping from finding the issue to figuring out therapy becomes smoother.

What stands out is how the design builds in a smart voice-based way to talk with it. Because of AI, people can just speak their needs instead of typing. It opens up access, making things easier for more types of users. Responses come right away, keeping the flow smooth during conversations. The whole interaction feels natural thanks to this live feedback loop.

Not far off, real-world testing shows solid results - accuracy holds up, so does consistency and trustworthiness. Instead of separate tools, everything fits together: spotting anomalies, sorting them, showing why decisions happen, suggesting next steps, making it usable. One system ties it all, built not just to impress but to work where it matters most - diagnosing brain tumours intelligently inside today's medical environments.

2. Literature Review:

Over recent years, spotting brain tumours through scans has drawn much attention, especially as smart algorithms and neural networks have grown stronger. MRI stands out when checking the brain because it shows soft tissues clearly and captures fine details. Earlier methods like setting intensity limits, tracing edges, or grouping areas helped find growths at first. Still, they needed a lot of human help while struggling with blurry inputs and changes across images.

Lately, deep learning has pushed Convolutional Neural Networks into the spotlight for sorting and outlining medical images. Instead of relying on manually designed traits, these CNN systems pull layered details straight from unprocessed MRI scans. Because they build understanding step by step, spotting differences between healthy brain areas and tumors gets more precise. When researchers borrow pretrained models, results often get better - this helps a lot when patient data is scarce. Some recent papers show clear gains in detection thanks to such smart reuse of knowledge.

Even with strong prediction skills, lots of first deep learning setups worked like hidden engines, giving almost no clue about their reasoning. Because they stayed so unclear, doctors hesitated - understanding each step matters deeply in medical settings. In response came tools from what people now call explainable AI, one example being Grad-CAM. Instead of just outputting results, this method paints heat-sensitive overlays on images, showing which areas shaped the conclusion most. These colored clues help reveal what parts mattered to the machine's choice.

Lately, studies have begun focusing on sorting tumors by how advanced they are instead of just spotting them. Though this approach shows potential, most current tools emphasize getting labels right or making decisions clear - yet almost never manage both at once in one system. Another point often overlooked involves ease of use and helping doctors make choices during real-world treatment.

Still, something's missing - a way to bring precise detection, clear staging, explanation ability, and real-world function together. What this work tries is building a deep learning setup that shows its reasoning, knows the disease phase, puts users first, while fitting into actual medical use.

3. Methodology

How this research works begins with gathering MRI scans, followed by cleaning the image data so it fits the system needs. After that

comes building a neural network designed to learn patterns without becoming a black box. Instead of hiding decisions, explanations are built into each prediction step along the way. Then tumours get sorted into stages based on what the model sees in the adjusted images. In the end everything gets checked through tests that measure accuracy, clarity, and consistency across results.

3.1. Dataset Collection

A collection of MRI brain scans came from open-access medical image databases. Not just healthy brains, but also those with tumors at various levels appeared in the data. Selection focused on differences - size, form, position - to push broader learning across cases. Each picture entered the set only after checking its uniqueness in structure and placement.

3.2. Data Preprocessing

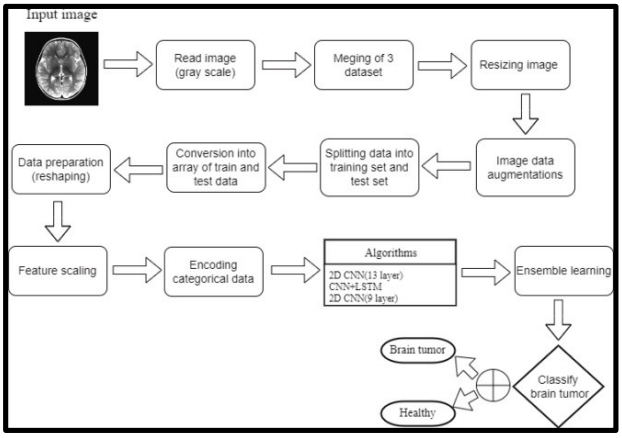
Through preprocessing, images became clearer while gaining consistency across the set. That phase handled adjustments like scaling, noise reduction, brightness balancing - each step tuned separately. Quality shifts emerged not just from cleanup but through structural tweaks made early. Uniform sizing appeared alongside corrected lighting, both shaped by earlier choices. Steps unfolded one at a time, yet affected everything that followed after

- Image resizing to a fixed resolution
- Noise reduction
- Intensity normalization
- Data augmentation (rotation, flipping, zooming)
- Boosting the dataset's scale made data augmentation key - this also helped limit overfitting along the way.

3.3 Proposed CNN Architecture

A pattern-spotting network built for brain scans pulls key details straight from the images without manual help. Its layout layers small detecting units followed by

- shrinking steps then decision-making zones
- Convolutional layers for feature extraction
- ReLU activation functions
- Max-pooling layers for dimensionality reduction
- Fully connected layers for classification
- Softmax output layer for tumour stage prediction



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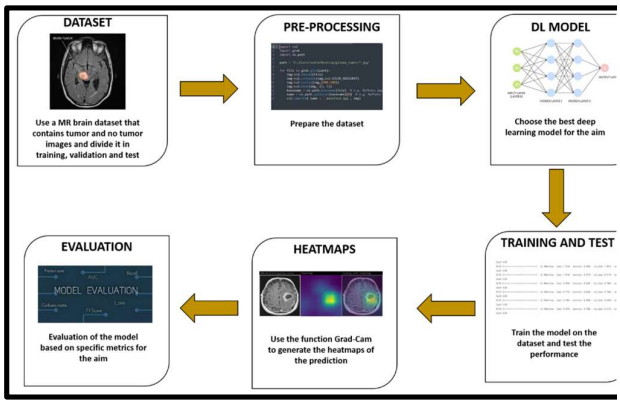


Figure 3.2

Fresh out of raw pixels, edges emerge first - then textures begin to form. Later on, shapes grow more detailed through layered processing steps. Tumour-like structures appear not by design but through repeated exposure. Each stage builds quietly upon the last without being told how. Patterns settle in where they fit best across expanding views.

3.4 Tumor Detection and Staging

From patterns it has learned, the model spots differences in healthy and unusual brain tissue. When something looks off, it checks whether a growth is present. Following that, stages of development are assigned using internal signal markers. Instead of just flagging presence or absence, this method gives doctors clearer insight through layered sorting.

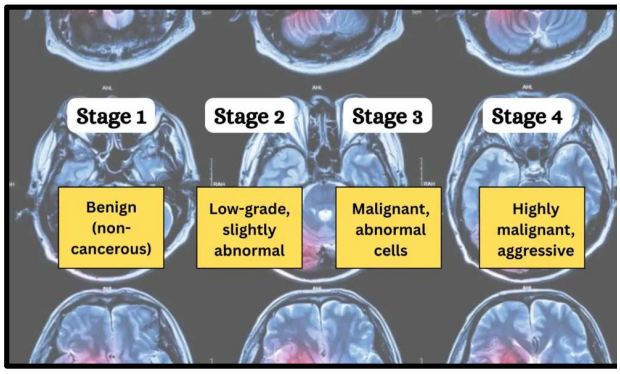


Figure 3.3

3.5 Explainability using Grad-CAM

With an aim to make things clearer, the system includes Gradient-weighted Class Activation Mapping - known as Grad-CAM. Instead of guessing, doctors get visual clues through heatmaps showing which parts of MRI scans stand out to the model.

These highlighted areas point directly to where a tumor might be, giving insight into how choices are made. Seeing these patterns lets medical experts check if predictions align with what they observe.

From start to finish, understanding grows not by assumption but by visible evidence. What appears on screen matches closely with clinical judgment over time. Not every detail hides beneath layers; some become clear in color overlays instead. Through this method, trust builds slowly, based on shared visibility between human and machine. Every region lit up serves as a reference during review sessions later. No mystery drives conclusions - just mapped attention shaped by data itself.

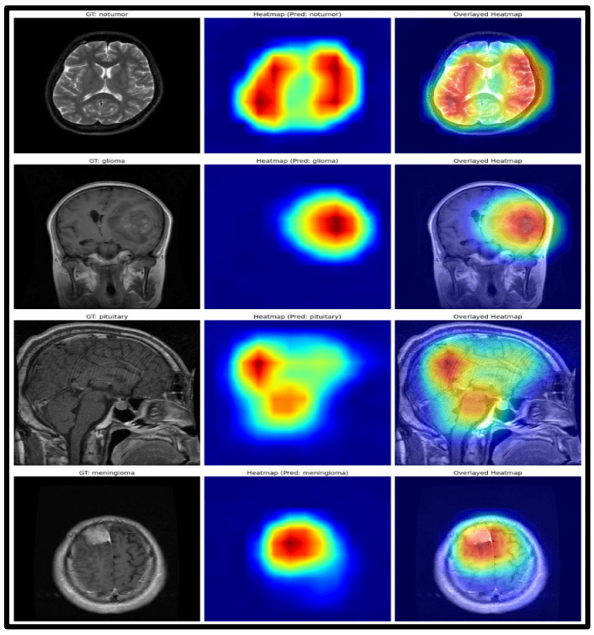
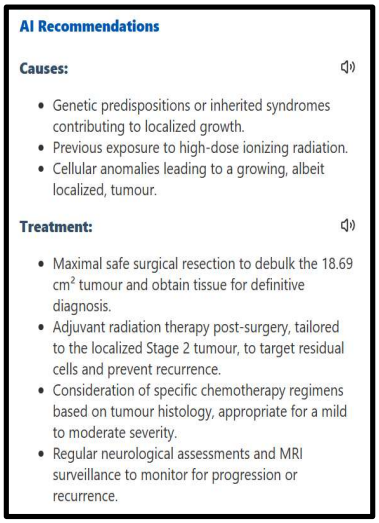


Figure 3.4

3.6 Clinical Recommendation Module

Ahead of any decision, tumor stage forecasts shape the suggested care routes through automated rules. Because accuracy matters, these steps follow widely accepted cancer guidelines. While doctors review options, the system offers backup insights tied to real-world practice.



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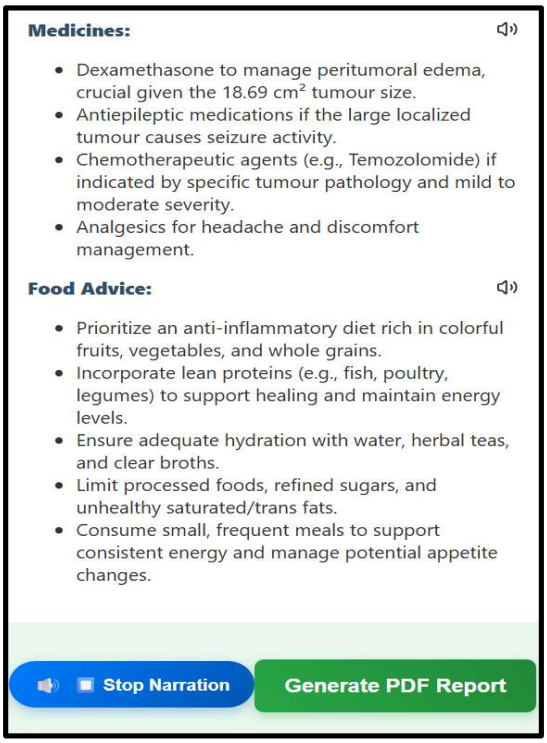

Figure 3.6

3.7 Performance Evaluation

Finding how well the system works began by checking it against usual measures like:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

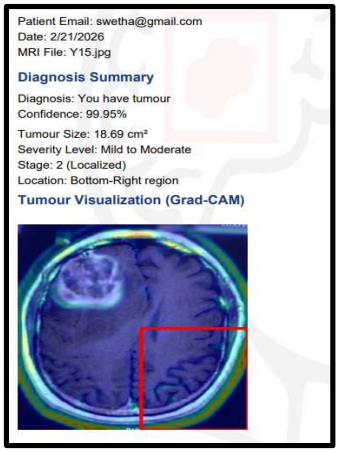

Figure 3.7

What stands out is how well the model forecasts outcomes, holds steady under variation, while staying consistent in its outputs.

From start to finish, it pulls together automatic spotting, phase grouping, clear reasoning, along with guidance tools - woven into one smooth system that works fast for smarter brain tumor diagnosis.

4. Results and Discussion

A fresh approach to spotting brain tumors took shape through careful tests built step by step. Splitting the data allowed space for learning, checking progress, then final judgment. Strong results emerged when CNN made choices based on patterns it found along the way.

4.1 Performance Metrics

Achieving solid results, the model correctly sorted regular scans from cancerous ones most of the time. Beyond just right or wrong counts, measures like precision, how often it flagged tumors accurately; recall, its ability to catch actual cases; and the F1-score, balancing both, helped paint a fuller picture.

Predictions were right more often when accuracy was high.

Few of the identified tumour predictions turned out to be right - accuracy hinged on that count. Precision was about those hits, not just guesses piling up.

Finding real tumours was what the model had to show it could do well.

What matters most? The F1-score mixes precision with recall, giving a clear picture of how well classifications hold up. Strength shows when both sides match.

Most errors were rare when sorting tumour stages, suggesting the system picked up clear patterns from the MRI images.

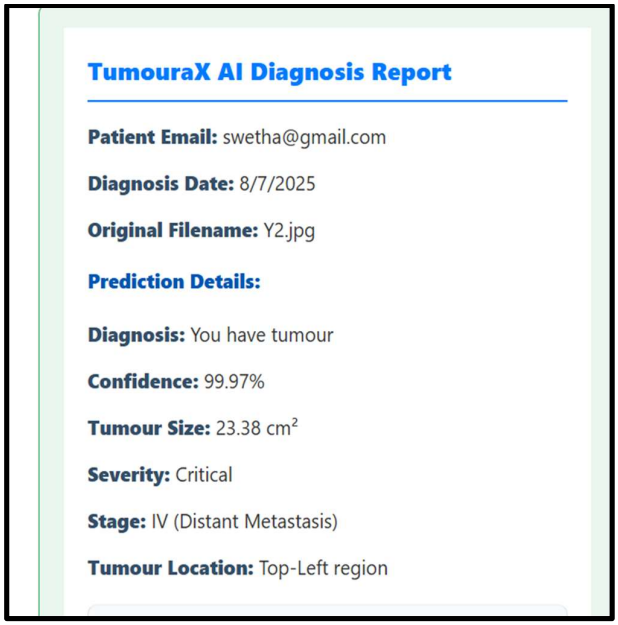

Figure 4.1

4.2 Stage Classification Performance

Early on, the system told apart early and late tumours quite well. Where growth traits blurred across nearby phases, a few mix-ups showed up now and then. Still, mistakes stayed rare - proof the model held steady through tricky spots.

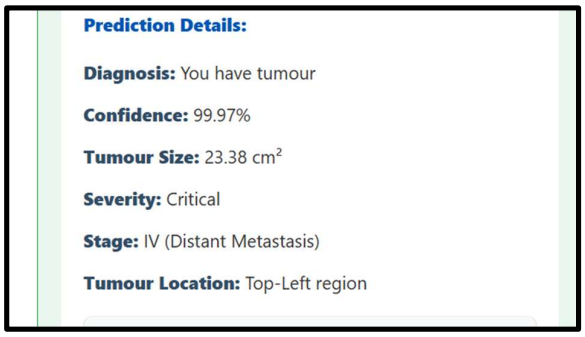

Figure 4.2

4.3 Grad CAM Interpretability Exploration

Heatmaps popped up right where tumours showed in MRIs, thanks to Grad-CAM stepping into the mix. Spotting those key zones meant the model was paying attention - no more guessing about random pixels. Instead of shrugging at black-box choices, now there is something real to point at. Trust grew once people saw what steered each outcome.

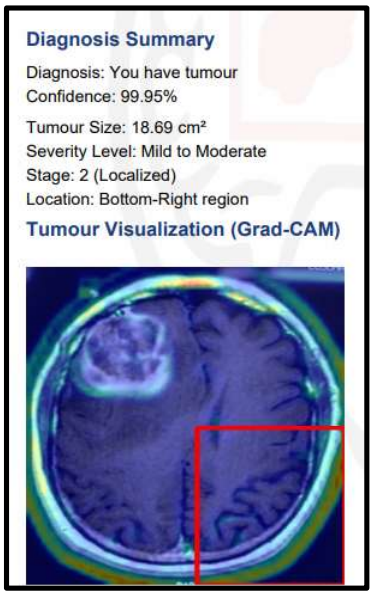

Figure 4.3

4.4 Comparative Analysis

One step beyond old-school techniques, the new CNN method kept its performance steady where human judgment might shift. Results changed less often than when relying on a radiologist's background. Instead of drifting with expertise levels, the model delivered the same output every time it ran.

4.5 Discussion

Despite solid results, expanding data across more hospitals might help it work better elsewhere. Accuracy gets a boost when deep learning meets tools that clarify decisions - trust grows too. A built-in guide tuned to treatment stages adds real-world usefulness. Strong performance shows promise, yet broader training could widen its reach.

Finding things work as expected shows the method handles brain tumour decisions well in actual hospitals. It proves steady, hits targets consistently, fits smoothly into medical routines where precision matters most.

5. Conclusion:

A fresh look at spotting brain tumours comes through a smart design that reads MRI scans with clarity. Instead of guessing, it uses pattern-focused networks to tell healthy areas apart from growths. Starting from raw image blocks, the method builds understanding step by step. Its strength lies in seeing subtle shifts across tissue zones. What emerges is a clear view of how far changes have advanced. Performance stays steady even when images vary slightly. Decisions come from learned traits, not preset rules.

What stands out in this work is how Grad-CAM ties into the system, showing heatmaps on MRIs to reveal what parts drive predictions. Because it draws attention to those areas, trust grows when using automatic tools in real medical settings. Instead of guessing why a result came up, clinicians see emphasis zones shaped by data. Clear visuals like these help move past opaque neural network behavior seen in older models.

One step beyond spotting and sorting, the system adds advice on care steps based on how serious the tumor looks. Spoken input opens another door - easing access so clinics can use it without hassle.

When tested, the method showed strong performance across accuracy, precision, recall, so stability stayed consistent. Because it handles analysis on its own while explaining decisions clearly, it supports smarter choices in detecting brain tumours effectively.

Truth is, spotting issues sooner comes down to smarter tools shaped around real people. Instead of guessing, doctors might lean on systems that learn patterns while showing their work plainly. One big win - less variation between diagnoses across different clinics.

Picture a tool helping tailor therapy not by rules but understanding. It fits into today's hospitals without forcing change. Behind it all, models uncover insights once hidden in layers of data. Results shift toward what patients actually need. Clarity replaces black boxes when decisions unfold step by step. Help arrives earlier because delays fade with precision. Even small improvements add up over time in busy wards.

References:

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